Anatomy

The body wall of the earthworm is very moist, thin, soft, skinny, elastic and consists of the cuticle, epidermis, muscles and coelomic epithelium. The epidermis consists of supporting cells, gland cells, basal cells and sensory cells. A spacious body cavity called the **coelom** is seen between the alimentary canal and the body wall. The coelom contains the coelomic fluid and serves as a **hydrostatic skeleton,** in which the coelomocytes are known to play a major role in regeneration, immunity and wound healing. The coelomic fluid of the earthworm is milky and alkaline, which consists of granulocytes or eleocytes, amoebocytes, mucocytes and leucocytes.

Digestive system

The digestive system of the earthworm consists of the alimentary canal and the digestive glands. The alimentary canal runs as a straight tube throughout the length of the body from the mouth to anus (Figure 4. 3).

The **mouth** opens into the **buccal cavity** which occupies the 1st and 2nd segments. The buccal cavity leads into a thick **muscular pharynx**, which occupies the 3rd and 4th segments and is surrounded by the pharyngeal glands. A small narrow tube, **oesophagus** lies in the 5th segment and continues into a muscular **gizzard** in the 6th segment. The gizzard helps in the grinding of soil particles and decaying leaves. **Intestine** starts from the 7th segment and continues till the last segment. The dorsal wall of the intestine is folded into the cavity as the **typhlosole**. This fold contains blood vessels and increases the absorptive area of the intestine. The inner epithelium consists

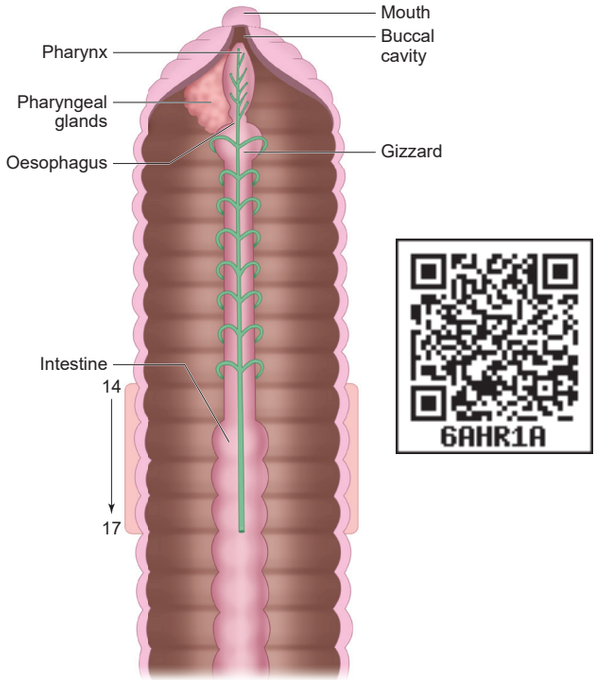


Figure 4.3 *Lampito mauritii* – Digestive System

of columnar cells and glandular cells. The alimentary canal opens to the exterior through the anus.

The ingested organic rich soil passes through the digestive tract where digestive enzymes breakdown complex food into smaller absorbable units. The simpler molecules are absorbed through the intestinal membrane and are utilized. The undigested particles along with earth are passed out through the anus, as **worm castings** or **vermicasts**. The pharyngeal

Intestinal Caeca

In *Metaphire posthuma*, the 26th segment has a pair of short conical out growths called intestinal caecae. It is extended anteriorly up to the 22nd segment. These are digestive glands and secrete an amylolytic enzyme for the digestion of starch. Intestinal caecae are not present in many species of earthworms such as the *Lampito mauritii*.

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